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Machine Learning-Based Error Detection in Networks

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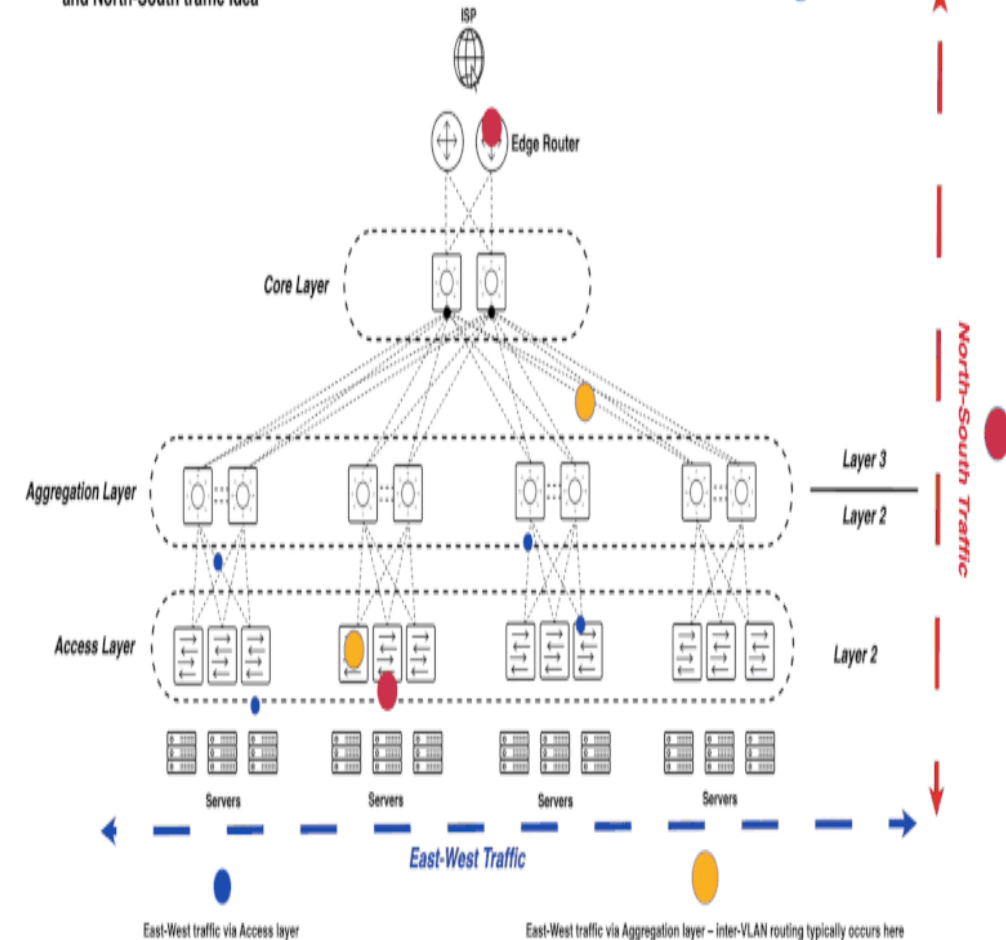
CSA0702: Computer Networks

Saveetha School of Engineering

INTRODUCTION

- Modern computer networks handle millions of data packets every second.
- Increasing internet users, cloud computing, IoT devices, and video streaming create heavy network traffic.
- Traditional traffic management techniques rely on static rules and struggle to adapt to dynamic traffic conditions.
- Artificial Intelligence (AI) enables intelligent traffic monitoring, prediction, and optimization to improve network performance.

3 tier architecture design East-West and North-South traffic Idea



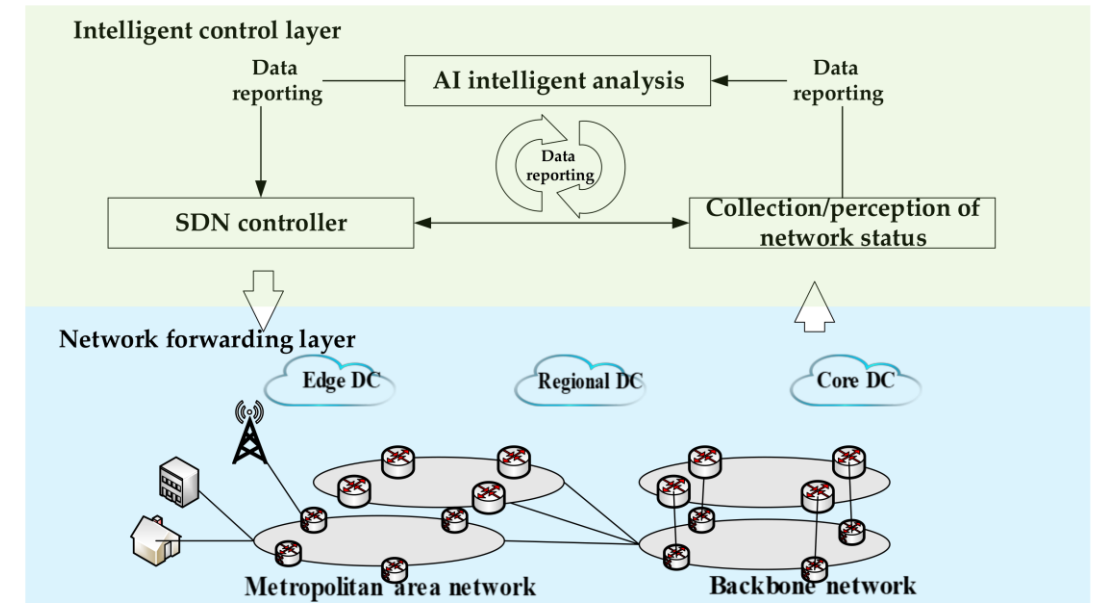
PROBLEM STATEMENT

Traditional network traffic management faces several challenges:

- Network congestion
- Packet loss
- High latency
- Poor bandwidth utilization
- Slow manual monitoring
- Difficulty predicting future traffic

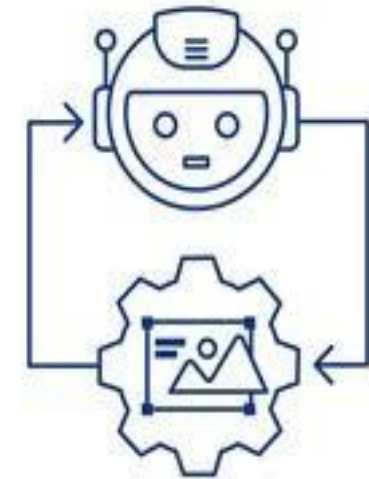
Proposed Solution

Use AI models to automatically analyze traffic, predict congestion, and optimize routing decisions.



OBJECTIVES

- Monitor network traffic continuously.
- Analyze traffic using AI algorithms.
- Predict congestion before it occurs.
- Optimize bandwidth utilization.
- Improve Quality of Service (QoS).
- Reduce packet loss and delay.
- Provide intelligent routing recommendations.



AI-Driven Workflow

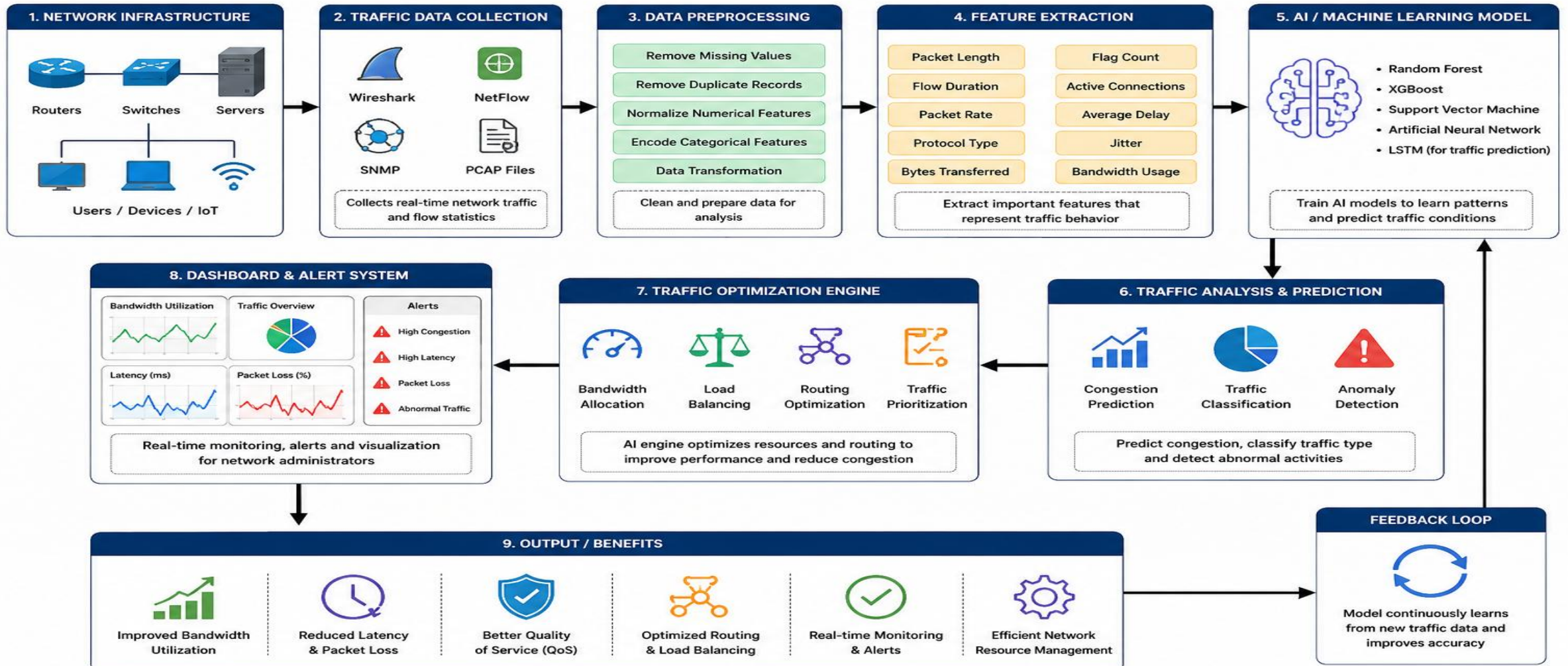


ENGINEER TO EXCEL



SYSTEM ARCHITECTURE

SYSTEM ARCHITECTURE ARTIFICIAL INTELLIGENCE FOR NETWORK TRAFFIC MANAGEMENT



CONCLUSION

Artificial Intelligence improves network traffic management by analyzing traffic patterns, predicting congestion, optimizing bandwidth allocation, and enhancing routing decisions. Compared to traditional approaches, AI-based systems provide higher efficiency, better Quality of Service, reduced latency, and improved scalability, making them suitable for modern cloud, IoT, enterprise, and next-generation communication networks.

THANK YOU!